

Lesson-01

Ex Chap-04, Pn2: A watermelon seed has the following coordinates: $x = -5.0 \text{ m}$, $y = 8.0 \text{ m}$, $z = 0 \text{ m}$. Find its position vector (a) in unit vector notation and as (b) a magnitude and (c) an angle relative to the positive direction of the x axis (d) Sketch the vector on a right-handed coordinate system. If the seed is moved to the xyz coordinates $(3.00 \text{ m}, 0 \text{ m}, 0 \text{ m})$, what is its displacement (e) in unit vector notation and as (f) a magnitude and (g) an angle relative to the positive x direction?

(a) The position vector according to eqⁿ 4-4 is

$$\vec{r} = (-5.0 \text{ m})\hat{i} + (8.0 \text{ m})\hat{j}$$

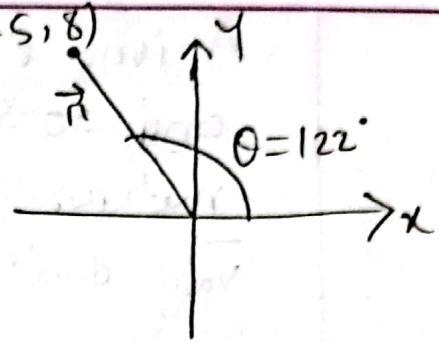
(b) The magnitude is

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$
$$= \sqrt{(-5)^2 + (8)^2}$$
$$|\vec{r}| = 9.4 \text{ m}$$

(c) $\theta = \tan^{-1}\left(\frac{8}{-5}\right) = -58^\circ \text{ or } 122^\circ$
= 122° angle is measured counterclockwise from the $+$ direction is chosen

Since the signs of the components imply the vector is in the second quadrant.

(d) The sketch is shown to the right. The vector 122° counterclockwise from the $+x$ direction.



(e) the displacement is

$$\begin{aligned}
 \Delta \vec{r} &= \vec{r}' - \vec{r} \\
 &= (-5\hat{i} + 8\hat{j}) - (3\hat{i} + 0\hat{j}) \\
 &= 3\hat{i} - (-5\hat{i} + 8\hat{j}) \\
 &= 8\hat{i} - 8\hat{j}
 \end{aligned}$$

(f) The magnitude of the displacement is

$$\begin{aligned}
 |\Delta \vec{r}| &= \sqrt{(8)^2 + (-8)^2} \\
 &= 11 \text{ m}
 \end{aligned}$$



(g) The angle of the displacement,

$$\theta = \tan^{-1}\left(\frac{-8}{8}\right)$$

$$= -45^\circ \text{ or } 135^\circ$$

where we choose the former possibility (-45° or 45° measured clockwise from $+x$) since the signs of the components imply the vector is in the fourth quadrant. A sketch of $\Delta \vec{r}$ is shown on the right.

chap:04 Problem:07:

A ion's position vector is initially $\vec{r} = 5\hat{i} - 6\hat{j} + 2\hat{k}$ and 10 s later it is $\vec{r} = -2\hat{i} + 8\hat{j} - 2\hat{k}$, all in meters. In unit vector notation, what is its $\vec{v}_{avg}$ during the 10 s?

Soln:

$$\vec{v}_{avg} = \frac{(-2\hat{i} + 8\hat{j} - 2\hat{k}) - (5\hat{i} - 6\hat{j} + 2\hat{k})}{10}$$
$$= -0.7\hat{i} + 1.4\hat{j} - 0.4\hat{k} \text{ m s}^{-1}$$

Problem:11 The position of a particle moving in an xy plane is given by $\vec{r} = (2t^3 - 5t)\hat{i} + (6 - 7t^4)\hat{j}$, with $\vec{r}$ in meters and t in seconds. In unit vector notation, calculate (a) $\vec{r}$, (b) $\vec{v}$ and (c) $\vec{a}$ for $t = 2.00 \text{ sec}$. (d) angle

Solution: (a) $\vec{r} = 2t^3\hat{i} - 5t\hat{i} + 6\hat{j} - 7t^4\hat{j}$

$$= 2.8\hat{i} - 5.2\hat{i} - 6\hat{j} - 7.16\hat{j}$$
$$= -2.4\hat{i} - 13.16\hat{j}$$

(b) $\vec{v} = \frac{d\vec{r}}{dt} = 6t^2\hat{i} - 5\hat{i} - 28t^3\hat{j}$

$$= 6 \cdot 2^2\hat{i} - 5\hat{i} - 28 \cdot 2^3\hat{j}$$
$$= 24\hat{i} - 5\hat{i} - 224\hat{j}$$
$$= 19\hat{i} - 224\hat{j}$$

$$(c) a = \frac{dv}{dt} = 12t \hat{i} - 84t^2 \hat{j} = 24 \hat{i} - 336 \hat{j} \text{ at } t=2.0$$

$$(d) \text{ The angle of } \vec{v}, \text{ measured from } +x, \text{ is either } \tan^{-1} \left(\frac{-224 \text{ m/s}}{19 \text{ m/s}} \right) = -85.2^\circ \text{ or } 94.8^\circ.$$

we settle on the first choice (-85.2° , which is equivalent to 275° measured counterclockwise from the $+x$ axis) since the signs of its components imply that it is in the fourth quadrant.

Problem 14: A proton initially has $\vec{v} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and the 4.0 s later has $\vec{v}' = -2\hat{i} - 2\hat{j} + 5\hat{k}$

For that 4.0 s what are (a) the proton's average acceleration $\vec{a}_{avg}$ in unit vector notation (b) the magnitude of $\vec{a}_{avg}$ and

(c) the angle betⁿ $\vec{a}_{avg}$ and the positive direction of the x axis?

(a) The average acceleration,

$$\vec{a}_{avg} = \frac{\vec{v}' - \vec{v}}{\Delta t} = \frac{(-2\hat{i} - 2\hat{j} + 5\hat{k}) - (4\hat{i} - 2\hat{j} + 3\hat{k})}{4}$$

$$= \frac{-6\hat{i} + 2\hat{k}}{4}$$

$$\vec{a}_{avg} = -1.5\hat{i} + 0.5\hat{k}$$

(b) The magnitude of $\vec{a}_{avg} = \sqrt{(-1.5 \text{ m/s}^2)^2 + (0.5)^2}$

$$\therefore \vec{a}_{avg} = 1.6 \text{ m/s}^2$$

(c) Its angle in the xz plane (measured from $+x$ axis)

$$\theta = \tan^{-1}\left(\frac{0.5}{-1.5}\right) = -18^\circ \text{ or } 162^\circ$$

where we settle on the second choice since the sign of its components imply that it is in the second quadrant.

Lesson 203

Chap: 04: Problem 21:

A dart is thrown horizontally with an initial speed 10 m/s^{-1} toward point P the bull's eye on a dart board. It hits at point Q on the rim, vertically below P, 0.19 s later. (a) what is the distance PQ? (b) How far away from the dart board is the dart.

Sol:

the initial velocity is horizontal so that $v_{0y} = 0$
and $v_{0x} = v_0 = 10 \text{ ms}^{-1}$

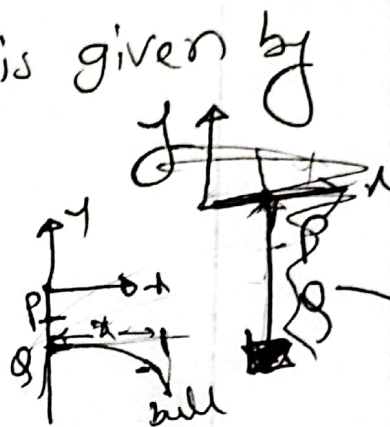
(a) The y co-ordinate of the dart is given by

$$y = -\frac{1}{2}gt^2 \quad [v_{0y} = 0]$$

$$\Rightarrow -1.8 = -\frac{1}{2}gt^2$$

$$1.8 = \frac{1}{2} \times 9.8 \times (0.19)^2$$

$$\therefore 1.8 = 0.18 \text{ m}$$



(b) From, $x = v_0 t$,

$$= 10 \times 0.19$$

$$= 1.9 \text{ m}$$

Problem 24: In the 1991 world track and field

championships in Tokyo, Mike Powell jumped 8.95 m, breaking by a full 5 cm the 23-year long - jump record set by Bob Beamon.

Assume that Powell's speed on takeoff was 9.5 m/s (about equal to that of a sprinter) and that $g = 9.80 \text{ ms}^{-2}$ in Tokyo.

How much less was Powell's range than the max^m possible range for a particle launched at the same speed?

Q. Soln: For max^m horizontal range $\theta = 45^\circ$

The horizontal range,

Here, $V_0 = 9.50 \text{ m/s}$

$$R = \frac{V_0^2 \sin 2\theta_0}{g}$$

$$= \frac{V_0^2}{g}$$

$$= \frac{(9.5)^2}{9.8}$$

$$= 9.21 \text{ m}$$

The difference from $R_{\text{max}} = (9.21 - 8.95)$
 $= 0.259 \text{ m}$

Or: 0.259 m

Problem 29: A projectile's launch speed is five times its speed at max^m height. Find launch angle θ_0 .

Soln: At max^m height, $V_y = 0$, $V_x = V_{0x} = V$

$$V_0 = 5V$$

$$\text{We know, } V_{0x} = V_0 \cos \theta_0$$

$$V = V_0 \cos \theta_0$$

$$\Rightarrow V = 5V \cos \theta_0$$

$$\theta_0 = \cos^{-1}\left(\frac{1}{5}\right)$$

$$\therefore \theta_0 = 78.5^\circ$$

$$2.37 \text{ m} \downarrow \begin{array}{c} 1 \\ 7 \end{array} \frac{1}{(0.0) \text{ m/s}} 12 \text{ m}$$

Problem 36: During a tennis match, a player serves the ball at 23.6 ms^{-1} , with the center of the ball leaving the racquet horizontally 2.37 m above the court surface. The net is 12 m away and 0.90 m high. When the ball reaches the net (a) does the ball clear it and (b) what is the distance betⁿ the center of the ball and the top of the net? Suppose that, instead, the ball is served as before but now it leaves the racquet at 5.0° below horizontal. When the ball reaches the net (c) does the ball clear it and (d) what now is the distance betⁿ the center of the ball and the top of the net?

(a) We want to know how high the ball is above the court when it is at $x = 12 \text{ m}$.

First, we need the time to cover the 12 m distance.

$$x = v_0 \cos \theta_0 t \quad [\theta_0 = 0]$$

$$\therefore t = \frac{x}{v_0 \cos \theta_0} \quad [\theta_0 = 0]$$

$$= \frac{12}{23.6 \times \cos 0^\circ}$$

$$= 0.508 \text{ sec}$$

At time t moved, the ball is at a height (above the court) of

$$y - y_0 = v_0 \sin \theta_0 t - \frac{1}{2} g t^2$$

$$y = y_0 + v_0 \sin \theta_0 t - \frac{1}{2} g t^2$$

$$= 2.37 + 0 - 4.9 \times (0.508)^2 \quad y_0 = 2.37$$

$$= 2.37 - 1.26 \text{ m}$$

$$= 1.10 \text{ m}$$

(b) At $t = 0.508 \text{ s}$, the center of the ball is $(1.10 \text{ m} - 0.90 \text{ m}) = 0.20 \text{ m}$ above the net

$$(c) \quad \text{At } \theta = -5^\circ, \quad t = \frac{v_0 \sin \theta_0}{g} = 0.510 \text{ sec}$$

$$= \frac{12}{23.6 \cos(-5^\circ)} = 0.510 \text{ sec}$$

$$\text{and } y = y_0 + v_0 \sin \theta_0 t - \frac{1}{2} g t^2$$

$$= 2.37 + 23.6 \sin(-5^\circ) \times 0.510 - 4.9 \times (0.510)^2$$

$$= -0.040 \text{ m}$$

which clearly indicate that it cannot clear the net.

(d) The distance betⁿ the top of the net and the center of the ball at $t = 0.510 \text{ sec}$ is.

$$d = 0.90 \text{ m} - 0.040 \text{ m} \\ = 0.86 \text{ m}$$

Lesson - 04

Chapter: 05

Problem 08:

A 2.0 kg object is subjected to three forces that give it an acceleration $\vec{a} = -(8 \text{ m/s}^2)\hat{i} + (6 \text{ m/s}^2)\hat{j}$. If two of the three forces are, $\vec{F}_1 = (30 \text{ N})\hat{i} + (16 \text{ N})\hat{j}$ and $\vec{F}_2 = -(12.0 \text{ N})\hat{i} + (8.0 \text{ N})\hat{j}$.

$$F_3 = ?$$

Solⁿ

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\Rightarrow \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = m\vec{a}$$

$$\therefore \vec{F}_3 = m\vec{a} - \vec{F}_1 - \vec{F}_2$$

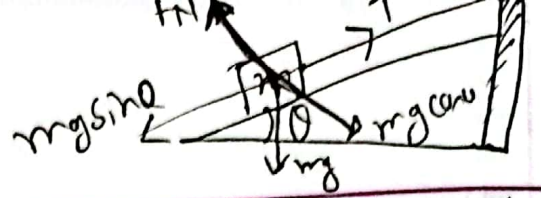
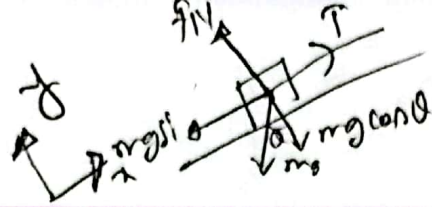
$$= 2 \text{ kg} \{ -8\hat{i} + 6\hat{j} \} - \{ 30\hat{i} + 16\hat{j} \} - \{ -12\hat{i} + 8\hat{j} \}$$

$$= -16\hat{i} + 12\hat{j} - 30\hat{i} - 16\hat{j} + 12\hat{i} - 8\hat{j}$$

$$\vec{F}_3 = -34\hat{i} - 12\hat{j}$$

Ans

1711



Let the mass of the block be 8.5 kg and the angle θ be 30° . Find (a) the tension in the cord and (b) the normal force acting on the block. (c) If the cord is cut, find the magnitude of the resulting acceleration of the block.

Sol: The free body diagram is shown below. Since the acceleration is zero, the Newton's 2nd law become:

$$T - mg \sin \theta = 0 \quad \text{①} \quad [\text{in } x\text{-direction}] \quad F_{\text{net},x} = ma_x$$

$$F_N - mg \cos \theta = 0 \quad \text{②} \quad [\text{in } y\text{-direction}] \quad F_{\text{net},y} = ma_y$$

(a) From eqn ① $\Rightarrow T = mg \sin \theta$

$$= 8.5 \times 9.8 \times \sin 30^\circ$$

$$\therefore T = 42 \text{ N}$$

(b) and eqn ② $\Rightarrow F_N = mg \cos \theta$

$$= 8.5 \times 9.8 \times \cos 30^\circ$$

$$= 72 \text{ N}$$

(c) When the cord is cut, the block is accelerated. The x components of the second law becomes

$$F_{\text{net}} = ma_x$$

$$\Rightarrow -mg \sin \theta = ma_x \quad \left[\begin{array}{l} \text{motion in} \\ \text{opposite dir} \\ \text{of } x \end{array} \right]$$

$$\begin{aligned}\Rightarrow a &= -g \sin \theta \\ &= -9.8 \times \sin 30^\circ \\ &= -4.9 \text{ m/s}^2\end{aligned}$$

The negative sign indicates the acceleration is down the plane. The magnitude of the acceleration is 4.9 m/s^2 .

25] A "Sun Yacht" is a spacecraft with a large sail that is pushed by sunlight. Although such a push is tiny in everyday circumstances, it can be large enough to send the spacecraft outward from the sun on a coast-free but slow trip. Suppose that the spacecraft has a mass of 900 kg and receives a push of 20 N. (a) what is the magnitude of the resulting acceleration? (b) If the craft starts from rest, how far will it travel in 1 day and (c) how fast will it then be moving?

solⁿ:

The acceleration,

$$a = \frac{F}{m} = \frac{20 \text{ N}}{900 \text{ kg}}$$

$$\therefore a = 0.022 \text{ m/s}^2$$

Here, $m = 900 \text{ kg}$

$F = 20 \text{ N}$

$a = ?$

$v_0 = 0$

$t = 1 \text{ day}$

$= 24 \times 3600$

$= 86400 \text{ s}$

$\therefore a = ?$

$v = ?$

(b) The distance traveled in 1 day = 86400 s

$$\begin{aligned}x &= v_0 t + \frac{1}{2} a t^2 \\&= \frac{1}{2} a t^2 \quad \left[v_0 = 0 \right] \\&= \frac{1}{2} \times 0.022 \times (86400)^2 \\ \therefore x &= 8.3 \times 10^7 \text{ m}\end{aligned}$$

(c) The speed, $v = v_0 + at$

$$= 0 + 0.022 \text{ m/s}^2 \times 86400 \text{ s}$$
$$\therefore v = 1.9 \times 10^3 \text{ ms}^{-1}$$

451 An elevator cab that weighs 27.8 kN moves upward. What is the tension in the cable if the cab's speed is (a) increasing at a rate of 1.22 ms^{-2} (b) decreasing at a rate of 1.22 ms^{-2}

Solⁿ:

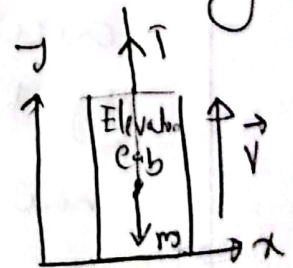
Here, $W = 27.8 \text{ kN}$

$$mg = 27.8 \text{ kN}$$

$$m = \frac{27.8 \times 10^3 \text{ N}}{9.8 \text{ ms}^{-2}}$$

$$= 2837 \text{ kg}$$

$$a = 1.22 \text{ ms}^{-2}$$



(a) Newton's second law leads to:

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\Rightarrow T - mg = ma$$

$$\therefore T = m(g + a)$$

$$= 2837(9.8 + 1.22)$$

$$T = 3.13 \times 10^4 \text{ N}$$

(b) Deceleration: Means the acceleration vector is in the opposite direction of velocity vector. Thus a is in the $-y$ direction.

$$\text{So, } a = -1.22 \text{ ms}^{-2}$$

$$\therefore T = m(g + a)$$

$$= 2837(9.8 - 1.22)$$

$$= 2.43 \times 10^4 \text{ N}$$